

LYRICA® CR
Pregabalin

1. NAME OF THE MEDICINAL PRODUCT

LYRICA® CR 82.5 and 165mg Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

LYRICA CR tablets contain 82.5 mg or 165 mg pregabalin.

3. PHARMACEUTICAL FORM

LYRICA CR 82.5 mg modified-release tablets: Light blue, film-coated, almond-shaped tablet debossed with “VLE” on 1 side and “PGN 82.5” on the other side.

LYRICA CR 165 mg modified-release tablets: Beige, film-coated, almond-shaped tablet debossed with “VLE” on 1 side and “PGN 165” on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

LYRICA CR is indicated for the management of:

- Neuropathic pain associated with diabetic peripheral neuropathy (DPN)
- Postherpetic neuralgia (PHN)

Efficacy of LYRICA CR has not been established for the management of fibromyalgia or generalised anxiety disorder or as adjunctive therapy for adult patients with partial onset seizures.

4.2 Posology and method of administration

LYRICA CR should be administered once daily after an evening meal.

LYRICA CR should be swallowed whole and should not be split, crushed, or chewed.

When discontinuing LYRICA CR, taper gradually over a minimum of 1 week.

Instruct patients that if they miss taking their dose of LYRICA CR after an evening meal, then they should take their usual dose of LYRICA CR prior to bedtime following a snack. If they miss taking the dose of LYRICA CR prior to bedtime, then they should take their usual dose of LYRICA CR following a morning meal. If they miss taking the dose of LYRICA CR following the morning meal, then they should take their usual dose of LYRICA CR at the usual time that evening following an evening meal.

Neuropathic pain associated with Diabetic Peripheral Neuropathy

Begin dosing at 165 mg once daily and increase to 330 mg once daily within 1 week based on individual patient response and tolerability. The maximum recommended dose of LYRICA CR is 330 mg once daily.

Although LYRICA was studied at 600 mg/day, there was no evidence that this dose conferred additional significant benefit and this dose was less well tolerated. In view of the dose-dependent adverse reactions with LYRICA, treatment with doses above 330 mg/day is not recommended for LYRICA CR.

Postherpetic Neuralgia

Begin dosing at 165 mg once daily and increase to 330 mg once daily within 1 week based on individual patient response and tolerability.

Patients who do not experience sufficient pain relief following 2 to 4 weeks of treatment with 330 mg once daily and who are able to tolerate LYRICA CR, may be treated with up to 660 mg once daily. In view of the dose-dependent adverse reactions and the higher rate of treatment discontinuation due to adverse reactions, dosing above 330 mg/day should be reserved only for those patients who have ongoing pain and are tolerating 330 mg daily. The maximum recommended dose of LYRICA CR is 660 mg once daily.

Conversion from LYRICA Capsules to LYRICA CR Modified Release Tablets

When switching from LYRICA Capsules to LYRICA CR, on the day of the switch, instruct patients to take their morning dose of LYRICA Capsules as prescribed and initiate LYRICA CR therapy after an evening meal.

Table 1. Conversion from LYRICA Capsules to LYRICA CR

LYRICA Capsules Total Daily Dose (dosed 2 or 3 times daily)^d	LYRICA CR Dose (dosed once a day)
75 mg/day	82.5 mg/day
150 mg/day	165 mg/day
225 mg/day	247.5 mg/day ^a
300 mg/day	330 mg/day
450 mg/day	495 mg/day ^b
600 mg/day	660 mg/day ^c

a. $247.5 \text{ mg} = 3 \times 82.5 \text{ mg}$ tablets taken once a day

b. $495 \text{ mg} = 3 \times 165 \text{ mg}$ tablets taken once a day

c. $660 \text{ mg} = 2 \times 330 \text{ mg}$ tablets taken once a day

d. Please see LYRICA Capsules

Discontinuation of pregabalin

If pregabalin has to be discontinued, it is recommended this should be done gradually over a minimum of 1 week.

Patients with renal impairment

Use of LYRICA CR is not recommended for patients with creatinine clearance (CL_{cr}) less than 30 mL/min or who are undergoing haemodialysis. Those patients should receive LYRICA Capsules.

In view of dose-dependent adverse reactions and since pregabalin is eliminated primarily by renal excretion, adjust the dose in patients with reduced renal function. Base the dose adjustment in patients with renal impairment on CL_{cr} , as indicated in Table 2. To use the dosing tables, an estimate of the patient's CL_{cr} in mL/min is needed. CL_{cr} in mL/min may be estimated from serum creatinine (mg/dL) determination using the Cockcroft and Gault equation:

$$CL_{cr} \text{ (mL/min)} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)} \times 0.85 \text{ for female patients}}{72 \times \text{serum creatinine (mg/dl)}}$$

Next, refer to the Dosage and Administration section to determine the recommended total daily dose based on indication, for a patient with normal renal function (CL_{cr} greater than or equal to 60 mL/min). Then refer to Table 2 to determine the corresponding renal adjusted dose.

(For example: A patient initiating LYRICA CR therapy for postherpetic neuralgia with normal renal function [CL_{cr} greater than or equal to 60 mL/min], receives a single daily dose of 165 mg/day pregabalin. Therefore, a renal impaired patient with a CL_{cr} of 50 mL/min would receive a single daily dose of 82.5 mg.)

Table 2. LYRICA CR Dosage Adjustment Based on Renal Function

Creatinine Clearance (CL_{cr}) (mL/min)	Total Pregabalin Daily Dose (mg/day)				Dose Regimen
greater than or equal to 60	165	330	495 ^a	660 ^b	Once a day
30–60	82.5	165	247.5 ^c	330	Once a day
less than 30/hemodialysis	Dose with LYRICA Capsules				

^a. 495 mg = 3 × 165 mg tablets taken once a day

^b. 660 mg = 2 × 330 mg tablets taken once a day

^c. 247.5 mg = 3 × 82.5 mg tablets taken once a day

Use in patients with hepatic impairment

No dosage adjustment is required for patients with hepatic impairment (see Section 5.2 Pharmacokinetic properties, Pharmacokinetics in special patient groups, Hepatic impairment).

Use in children and adolescents (12 to 17 years of age)

The safety and effectiveness of pregabalin in pediatric patients below the age of 12 years and adolescents has not been established.

The use in children is not recommended (see Section 5.3 Preclinical safety data).

Use in the elderly (over 65 years of age)

Elderly patients may require a dose reduction of pregabalin due to decreased renal function (see Section 5.2 Pharmacokinetic properties, Pharmacokinetics in special patient groups, Elderly (over 65 years of age)).

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients.

4.4 Special warnings and precautions for use

Angioedema

There have been post-marketing reports of angioedema in patients during initial and chronic treatment with pregabalin. Specific symptoms included swelling of the face, mouth (tongue, lips, and gums), and neck (throat and larynx). There were reports of life-threatening angioedema with respiratory

compromise requiring emergency treatment. Discontinue pregabalin immediately in patients with these symptoms.

Exercise caution when prescribing pregabalin to patients who have had a previous episode of angioedema. In addition, patients who are taking other drugs associated with angioedema (e.g., angiotensin converting enzyme inhibitors [ACE-inhibitors]) may be at increased risk of developing angioedema.

Hypersensitivity

There have been post-marketing reports of hypersensitivity in patients shortly after initiation of treatment with pregabalin. Adverse reactions included skin redness, blisters, hives, rash, dyspnea, and wheezing. Discontinue pregabalin immediately in patients with these symptoms.

Suicidal Behavior and Ideation

Antiepileptic drugs (AEDs), including pregabalin, increase the risk of suicidal thoughts or behavior in patients taking these drugs for any indication. Suicidal behavior and ideation have also been reported in patients after discontinuation of pregabalin (see Section **4.4 Special warnings and precautions for use**). Monitor patients treated with any AED for any indication for the emergence or worsening of depression, suicidal thoughts or behavior, and/or any unusual changes in mood or behavior.

Pooled analyses of 199 placebo-controlled clinical trials (mono- and adjunctive therapy) of 11 different AEDs showed that patients randomized to one of the AEDs had approximately twice the risk (adjusted Relative Risk 1.8, 95% CI:1.2, 2.7) of suicidal thinking or behavior compared to patients randomized to placebo. In these trials, which had a median treatment duration of 12 weeks, the estimated incidence rate of suicidal behavior or ideation among 27,863 AED-treated patients was 0.43%, compared to 0.24% among 16,029 placebo-treated patients, representing an increase of approximately one case of suicidal thinking or behavior for every 530 patients treated. There were four suicides in drug-treated patients in the trials and none in placebo-treated patients, but the number is too small to allow any conclusion about drug effect on suicide.

The increased risk of suicidal thoughts or behavior with AEDs was observed as early as one week after starting drug treatment with AEDs and persisted for the duration of treatment assessed. Because most trials included in the analysis did not extend beyond 24 weeks, the risk of suicidal thoughts or behavior beyond 24 weeks could not be assessed.

The risk of suicidal thoughts or behavior was generally consistent among drugs in the data analyzed. The finding of increased risk with AEDs of varying mechanisms of action and across a range of indications suggests that the risk applies to all AEDs used for any indication. The risk did not vary substantially by age (5-100 years) in the clinical trials analyzed.

Table 3 shows absolute and relative risk by indication for all evaluated AEDs.

Table 3. Risk by Indication for Antiepileptic Drugs in the Pooled Analysis

Indication	Placebo Patients with Events Per 1000 Patients	Drug Patients with Events Per 1000 Patients	Relative Risk: Incidence of Events in Drug Patients/Incidence in Placebo Patients	Risk Difference: Additional Drug Patients with Events Per 1000 Patients
Epilepsy	1.0	3.4	3.5	2.4
Psychiatric	5.7	8.5	1.5	2.9

Other	1.0	1.8	1.9	0.9
Total	2.4	4.3	1.8	1.9

The relative risk for suicidal thoughts or behavior was higher in clinical trials for epilepsy than in clinical trials for psychiatric or other conditions, but the absolute risk differences were similar for the epilepsy and psychiatric indications.

Anyone considering prescribing pregabalin or any other AED must balance the risk of suicidal thoughts or behavior with the risk of untreated illness. Epilepsy and many other illnesses for which AEDs are prescribed are themselves associated with morbidity and mortality and an increased risk of suicidal thoughts and behavior. Should suicidal thoughts and behavior emerge during treatment, the prescriber needs to consider whether the emergence of these symptoms in any given patient may be related to the illness being treated.

Respiratory Depression

There is evidence from case reports, human studies, and animal studies associating pregabalin with serious, life-threatening, or fatal respiratory depression when co-administered with central nervous system (CNS) depressants, including opioids, or in the setting of underlying respiratory impairment. When the decision is made to co-prescribe pregabalin with another CNS depressant, particularly an opioid, or to prescribe pregabalin to patients with underlying respiratory impairment, monitor patients for symptoms of respiratory depression and sedation, and consider initiating pregabalin at a low dose. The management of respiratory depression may include close observation, supportive measures, and reduction or withdrawal of CNS depressants (including pregabalin).

There is more limited evidence from case reports, animal studies, and human studies associating pregabalin with serious respiratory depression, without co-administered CNS depressants or without underlying respiratory impairment.

Dizziness and Somnolence

Pregabalin may cause dizziness and somnolence. Inform patients that pregabalin-related dizziness and somnolence may impair their ability to perform tasks such as driving or operating machinery and could increase the occurrence of accidental injury (fall) in the elderly population.

In the pregabalin controlled trials in adult patients, dizziness was experienced by 24% of pregabalin-treated patients compared to 8% of placebo-treated patients; somnolence was experienced by 15.8% of pregabalin-treated patients compared to 8% of placebo-treated patients. Dizziness and somnolence generally began shortly after the initiation of pregabalin therapy and occurred more frequently at higher doses. Dizziness and somnolence were the adverse reactions most frequently leading to withdrawal (1.2% each) from controlled studies. In pregabalin-treated patients reporting these adverse reactions in short-term, controlled studies, dizziness persisted until the last dose in 30% and somnolence persisted until the last dose in 42% of patients (see Section **4.5 Interaction with other medicinal products and other forms of interaction**).

Increased Risk of Adverse Reactions with Abrupt or Rapid Discontinuation

As with all antiepileptic drugs (AEDs), withdraw pregabalin gradually to minimize the potential of increased seizure frequency in patients with seizure disorders.

Following abrupt or rapid discontinuation of LYRICA CR, some patients reported symptoms including seizures, depression, suicidal ideation and behavior, agitation, confusion, disorientation, psychotic symptoms, anxiety, insomnia, nausea, pain, sweating, tremor, headache, malaise, and

diarrhea. Suicidal behavior and ideation have also been reported in patients after discontinuation of pregabalin (see Section 4.4 Special warnings and precautions for use).

If LYRICA CR is discontinued, taper the drug gradually over a minimum of 1 week rather than discontinue the drug abruptly.

The efficacy of LYRICA CR as adjunctive therapy for adult patients with partial onset seizures has not been established.

Peripheral Edema

Pregabalin treatment may cause peripheral edema. In short-term trials of patients without clinically significant heart or peripheral vascular disease, there was no apparent association between peripheral edema and cardiovascular complications such as hypertension or congestive heart failure. Peripheral edema was not associated with laboratory changes suggestive of deterioration in renal or hepatic function.

In controlled clinical trials for pain indications, the incidence of peripheral edema for patients receiving LYRICA CR in the single-blind phase was 5.3% of patients. In controlled clinical trials for pain indications, 0.8% of LYRICA CR patients withdrew due to peripheral edema during the single-blind phase.

Higher frequencies of weight gain and peripheral edema were observed in patients taking both pregabalin and a thiazolidinedione antidiabetic agent compared to patients taking either drug alone. The majority of patients using thiazolidinedione antidiabetic agents in the overall safety database were participants in studies of pain associated with diabetic peripheral neuropathy. In this population, peripheral edema was reported in 3% (2/60) of patients who were using thiazolidinedione antidiabetic agents only, 8% (69/859) of patients who were treated with pregabalin only, and 19% (23/120) of patients who were on both pregabalin and thiazolidinedione antidiabetic agents. Similarly, weight gain was reported in 0% (0/60) of patients on thiazolidinediones only; 4% (35/859) of patients on pregabalin only; and 7.5% (9/120) of patients on both drugs.

As the thiazolidinedione class of antidiabetic drugs can cause weight gain and/or fluid retention, possibly exacerbating or leading to heart failure, exercise caution when co-administering pregabalin and these agents.

Because there are limited data on congestive heart failure patients with New York Heart Association (NYHA) Class III or IV cardiac status, monitor these patients for possible exacerbation of congestive heart failure symptoms when using LYRICA CR.

Weight Gain

Pregabalin treatment may cause weight gain. In pregabalin controlled trials for pain indications, weight gain was experienced by 4% of LYRICA CR-treated patients during the single-blind phase. Adverse events of weight gain were observed in 3.7% of LYRICA CR-treated patients and 1% of placebo-treated patients during the double-blind phase.

In pregabalin controlled clinical trials in adult patients of up to 14 weeks, a gain of 7% or more over baseline weight was observed in 9% of pregabalin-treated patients and 2% of placebo-treated patients. Few patients treated with pregabalin (0.3%) withdrew from controlled trials due to weight gain. Pregabalin associated weight gain was related to dose and duration of exposure, but did not appear to be associated with baseline BMI, gender, or age. Weight gain was not limited to patients with edema (see Section 4.4 Special warnings and precautions for use).

Although weight gain was not associated with clinically important changes in blood pressure in short-term controlled studies, the long-term cardiovascular effects of pregabalin-associated weight gain are unknown.

Among diabetic patients, pregabalin-treated patients gained an average of 1.6 kg (range: -16 to 16 kg), compared to an average 0.3 kg (range: -10 to 9 kg) weight gain in placebo patients. In a cohort of 333 diabetic patients who received pregabalin for at least 2 years, the average weight gain was 5.2 kg.

While the effects of pregabalin-associated weight gain on glycemic control have not been systematically assessed, in controlled and longer-term open label clinical trials with diabetic patients, pregabalin treatment did not appear to be associated with loss of glycemic control (as measured by HbA_{1C}).

Tumorigenic Potential

In standard preclinical *in vivo* lifetime carcinogenicity studies of pregabalin, an unexpectedly high incidence of hemangiosarcoma was identified in two different strains of mice. The clinical significance of this finding is unknown. Clinical experience during pregabalin's pre-marketing development provides no direct means to assess its potential for inducing tumors in humans.

In clinical studies across various patient populations, comprising 6396 patient-years of exposure in patients greater than 12 years of age, new- or worsening-preexisting tumors were reported in 57 patients. Without knowledge of the background incidence and recurrence in similar populations not treated with pregabalin, it is impossible to know whether the incidence seen in these cohorts is or is not affected by treatment.

Ophthalmological Effects

In controlled studies for pain indications, 4.8% of patients treated with LYRICA CR in the single-blind phase reported blurred vision, which resolved in a majority of cases with continued dosing. Less than 1% of patients discontinued LYRICA CR treatment due to vision-related events (primarily blurred vision). Additionally, 0.7% of LYRICA CR-treated patients as compared to no placebo-treated patients experienced blurred vision in the double-blind phase.

Prospectively planned ophthalmologic testing during the premarketing development of pregabalin, including visual acuity testing, formal visual field testing and dilated funduscopy examination, was performed in over 3600 patients. In these patients, visual acuity was reduced in 7% of LYRICA-treated patients and 5% of placebo-treated patients. Visual field changes were detected in 13% of LYRICA-treated and 12% of placebo-treated patients. Funduscopy changes were observed in 2% of LYRICA-treated and 2% of placebo-treated patients.

Although the clinical significance of the ophthalmologic findings is unknown, inform patients to notify their physician if changes in vision occur. If visual disturbance persists, consider further assessment. Consider more frequent assessment for patients who are already routinely monitored for ocular conditions.

Creatine Kinase Elevations

Pregabalin treatment was associated with creatine kinase elevations. Mean changes in creatine kinase from baseline to the maximum value were 60 U/L for pregabalin-treated patients and 28 U/L for the placebo patients. In all controlled trials in adult patients across multiple patient populations, 1.5% of patients on pregabalin and 0.7% of placebo patients had a value of creatine kinase at least three times the upper limit of normal. Three pregabalin treated subjects had events reported as rhabdomyolysis in premarketing clinical trials. The relationship between these myopathy events and pregabalin is not completely understood because the cases had documented factors that may have caused or contributed to these events. Instruct patients to promptly report unexplained muscle pain, tenderness, or weakness,

particularly if these muscle symptoms are accompanied by malaise or fever. Discontinue treatment with pregabalin if myopathy is diagnosed or suspected or if markedly elevated creatine kinase levels occur.

Decreased Platelet Count

Pregabalin treatment was associated with a decrease in platelet count. In the double-blind phase of controlled studies for pain indication, LYRICA CR-treated patients experienced a median change from baseline in platelet count of $11 \times 10^3/\text{mm}^3$ (for the PHN population) and $14 \times 10^3/\text{mm}^3$ (for the FM population) as compared to $1 \times 10^3/\text{mm}^3$ in placebo-treated patients (for both populations). Pregabalin-treated subjects experienced a mean maximal decrease in platelet count of $20 \times 10^3/\mu\text{L}$, compared to $11 \times 10^3/\mu\text{L}$ in placebo patients. In the overall database of controlled trials in adult patients, 2% of placebo patients and 3% of pregabalin patients experienced a potentially clinically significant decrease in platelets, defined as 20% below baseline value and less than $150 \times 10^3/\mu\text{L}$. A single pregabalin treated subject developed severe thrombocytopenia with a platelet count less than $20 \times 10^3/\mu\text{L}$. In randomized controlled trials, pregabalin was not associated with an increase in bleeding-related adverse reactions.

PR Interval Prolongation

Pregabalin treatment was associated with PR interval prolongation. In analyses of clinical trial ECG data in adult patients, the mean PR interval increase was 3–6 msec at pregabalin doses greater than or equal to 300 mg/day. This mean change difference was not associated with an increased risk of PR increase greater than or equal to 25% from baseline, an increased percentage of subjects with on-treatment PR greater than 200 msec, or an increased risk of adverse reactions of second- or third-degree AV block.

Subgroup analyses did not identify an increased risk of PR prolongation in patients with baseline PR prolongation or in patients taking other PR prolonging medications. However, these analyses cannot be considered definitive because of the limited number of patients in these categories.

Women of childbearing potential/Contraception

Pregabalin use in the first trimester of pregnancy may cause major birth defects in the unborn child. Pregabalin should not be used during pregnancy unless the benefit to the mother clearly outweighs the potential risk to the fetus. Women of childbearing potential must use effective contraception during treatment (see Section **4.6 Fertility, pregnancy and lactation**).

Others

Although the effects of discontinuation on the reversibility of renal failure have not been systematically studied, improved renal failure following discontinuation or dose reduction of pregabalin has been reported.

Misuse, abuse potential or dependence

Cases of misuse, abuse and dependence have been reported. Caution should be exercised in patients with a history of substance abuse and the patient should be monitored for symptoms of pregabalin misuse, abuse or dependence (development of tolerance, dose escalation, drug seeking behaviour have been reported).

Encephalopathy

Cases of encephalopathy have been reported, mostly in patients with underlying conditions that may precipitate encephalopathy.

Lactose intolerance

Pregabalin contains lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, the Lapp-lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

Treatment of central neuropathic pain due to spinal cord injury

In the treatment of central neuropathic pain due to spinal cord injury the incidence of adverse reactions in general, central nervous system adverse reactions and especially somnolence was increased. This may be attributed to an additive effect due to concomitant medicinal products (e.g., anti-spasticity agents) needed for this condition. This should be considered when prescribing pregabalin in this condition.

4.5 Interaction with other medicinal products and other forms of interaction

Since pregabalin is predominantly excreted unchanged in the urine, undergoes negligible metabolism in humans (less than 2% of a dose recovered in urine as metabolites), and does not bind to plasma proteins, its pharmacokinetics are unlikely to be affected by other agents through metabolic interactions or protein binding displacement. In vitro studies showed that pregabalin is unlikely to be involved in significant pharmacokinetic drug interactions (see Section **5.2 Pharmacokinetic properties**).

The interactions of LYRICA CR with co-administration of other drugs have not been systematically evaluated. Co-administration of the prokinetic drug erythromycin with LYRICA CR did not result in any clinically important changes in the pharmacokinetics of LYRICA CR (see Section **5.2 Pharmacokinetic properties**).

Additional studies have been performed with LYRICA. No pharmacokinetic interactions were observed between LYRICA and carbamazepine, gabapentin, lamotrigine, oral contraceptive, phenobarbital, phenytoin, topiramate, and valproic acid. A similar lack of pharmacokinetic interactions would be expected to occur with LYRICA CR.

Pharmacodynamics

Although no pharmacokinetic interactions were seen with LYRICA and ethanol, lorazepam, or oxycodone, additive effects on cognitive and gross motor functioning were seen when LYRICA was co-administered with these drugs. No clinically important effects on respiration were seen in studies of LYRICA.

4.6 Fertility, pregnancy and lactation

There is a limited amount of data on the use of pregabalin in pregnant women.

Data from an observational study, which included more than 2,700 pregnancies exposed to pregabalin based on routinely collected data from administrative and medical registers in Denmark, Finland, Norway, and Sweden, do not suggest substantially increased risks of major congenital malformations, adverse birth outcomes, or abnormal postnatal neurodevelopmental outcomes in pregabalin-exposed pregnancies.

Major congenital malformations

The adjusted prevalence ratios (aPRs) and 95% confidence intervals (CI) in the standard meta-analysis for first-trimester pregabalin monotherapy-exposed vs. unexposed to anti-epileptic drugs was 1.14 (0.96-1.35).

Birth and postnatal neurodevelopmental outcomes

There were no statistically significant findings for stillbirth, low birth weight, preterm birth, small for gestational age, low Apgar score, and microcephaly.

In pediatric population exposed *in utero*, the study did not provide evidence of an increased risk for attention deficit hyperactivity disorder (ADHD), autism spectrum disorders (ASD), and intellectual disabilities.

Studies in animals have shown reproductive toxicity (see Section **5.3 Preclinical safety data**). Pregabalin should not be used during pregnancy unless the benefit to the mother clearly outweighs the potential risk to the foetus. Effective contraception must be used in women of child bearing potential.

Lactation

Pregabalin is excreted in the milk of lactating women (see Section **5.2 Pharmacokinetic properties**). As the safety of pregabalin in infants is not known, breast-feeding is not recommended during treatment with pregabalin. A decision must be made whether to discontinue breast-feeding or to discontinue from pregabalin therapy taking into account the benefit of breast feeding for the child and the benefit of therapy for the woman.

4.7 Effects on ability to drive and use machines

Pregabalin may cause dizziness and somnolence and therefore may influence the ability to drive or use machines. Patients are advised not to drive, operate complex machinery or engage in other potentially hazardous activities until it is known whether this medication affects their ability to perform these activities.

4.8 Undesirable effects

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

Two randomized placebo-controlled clinical trials were conducted in patients with postherpetic neuralgia and fibromyalgia in which a total of 1242 patients received LYRICA CR. Both studies were randomized withdrawal design where a 6-week single-blind, dose optimization phase was followed by a 13-week double-blind phase. The most common adverse events leading to discontinuation from the single-blind phase of the study occurring in greater than or equal to 0.3% of patients were dizziness, somnolence, peripheral edema, fatigue, blurred vision, and increased weight. Sixty-four percent of patients experienced adverse events during the single-blind phase, with the most common adverse events occurring in greater than or equal to 4% of patients being dizziness, somnolence, headache, fatigue, peripheral edema, nausea, blurred vision, dry mouth, and weight gain.

The pregabalin clinical program involved over 12,000 patients who were exposed to pregabalin, of whom over 7,000 were in double-blind placebo controlled trials. The most commonly reported adverse reactions were dizziness and somnolence. Adverse reactions were usually mild to moderate in intensity. In all controlled studies, the discontinuation rate due to adverse reactions was 14% for patients receiving pregabalin and 5% for patients receiving placebo. The most common adverse reactions resulting in discontinuation from pregabalin treatment groups were dizziness and somnolence.

Selected adverse drug reactions that were treatment related in the pooled analysis of clinical trials are listed in the table below by System Organ Class (SOC). The frequency of these terms has been based on all-causality adverse drug reactions in the clinical trial data set (very common ($\geq 1/10$), common ($\geq 1/100$, $< 1/10$), uncommon ($\geq 1/1000$, $< 1/100$) and rare ($< 1/1000$)).

The adverse reactions listed may also be associated with the underlying disease and/or concomitant medications.

Table 4. Adverse Drug Reactions from Clinical Trial Experience

System Organ Class	Adverse Drug Reactions
Infections and infestations	
Common	Nasopharyngitis
Blood and lymphatic system disorders	
Uncommon	Neutropenia
Metabolism and nutrition disorders	
Common	Appetite increased
Uncommon	Anorexia, hypoglycemia
Psychiatric disorders	
Common	Euphoric mood, confusion, irritability, depression, disorientation, insomnia, libido decreased
Uncommon	Hallucination, restlessness, agitation, depressed mood, elevated mood, mood swings, depersonalization, abnormal dreams, word finding difficulty, libido increased, anorgasmia
Rare	Panic attack, disinhibition, apathy
Nervous system disorders	
Very Common	Dizziness, somnolence
Common	Ataxia, coordination abnormal, tremor, dysarthria, amnesia, memory impairment, disturbance in attention, paraesthesia, hypoesthesia, sedation, balance disorder, lethargy
Uncommon	Syncope, myoclonus, psychomotor hyperactivity, dyskinesia, dizziness postural, intention tremor, nystagmus, cognitive disorder, speech disorder, hyporeflexia, hyperaesthesia, burning sensation
Rare	Stupor, parosmia, hypokinesia, ageusia, dysgraphia
Eye disorders	
Common	Vision blurred, diplopia
Uncommon	Peripheral vision loss, visual disturbance, eye swelling, visual field defect, visual acuity reduced, eye pain, asthenopia, photopsia, dry eye, lacrimation increased, eye irritation
Rare	Oscillopsia, altered visual depth perception, mydriasis, strabismus, visual brightness
Ear and labyrinth disorders	
Common	Vertigo
Uncommon	Hyperacusis

Cardiac disorders	
Uncommon	Tachycardia, atrioventricular block first degree, sinus bradycardia
Rare	Sinus tachycardia, sinus arrhythmia
Vascular disorders	
Uncommon	Hypotension, hypertension, hot flushes, flushing, peripheral coldness
Respiratory, thoracic and mediastinal disorders	
Uncommon	Dyspnoea, epistaxis, cough, nasal congestion, rhinitis, snoring
Rare	Throat tightness, nasal dryness
Gastrointestinal disorders	
Common	Vomiting, constipation, flatulence, abdominal distension, dry mouth
Uncommon	Gastroesophageal reflux disease, salivary hypersecretion, hypoesthesia oral
Rare	Ascites, pancreatitis, dysphagia
Skin and subcutaneous tissue disorders	
Uncommon	Rash papular, urticaria, sweating
Rare	Cold sweat
Musculoskeletal and connective tissue disorders	
Common	Muscle cramp, arthralgia, back pain, pain in limb, cervical spasm
Uncommon	Joint swelling, myalgia, muscle twitching, neck pain, muscle stiffness
Rare	Rhabdomyolysis
Renal and urinary disorders	
Uncommon	Urinary incontinence, dysuria
Rare	Renal failure, oliguria
Reproductive system and breast disorders	
Uncommon	Erectile dysfunction, sexual dysfunction, ejaculation delayed, dysmenorrhoea
Rare	Breast pain, amenorrhoea, breast discharge, breast enlargement
General disorders and administration site conditions	
Common	Oedema peripheral, oedema, gait abnormal, fall, feeling drunk, feeling abnormal, fatigue
Uncommon	Generalised oedema, chest tightness, pain, pyrexia, thirst, chills, asthenia
Investigations	
Common	Weight increased
Uncommon	Blood creatine phosphokinase increased, alanine aminotransferase increased, aspartate aminotransferase increased, blood glucose increased, platelet count decreased, blood potassium decreased, weight decreased
Rare	White blood cell count decreased, blood creatinine increased

The following adverse drug reactions were reported during POST-MARKETING SURVEILLANCE:

Immune system disorder: Uncommon: Hypersensitivity; Rare: Angioedema, allergic reaction.

Nervous system disorders: Very Common: Headache; Uncommon: Loss of consciousness, mental impairment.

Eye disorders: Rare: Keratitis.§

Cardiac disorders: Rare: Congestive heart failure.

Respiratory, thoracic and mediastinal disorders: Rare: Pulmonary oedema.§

Gastrointestinal disorders: Common: Nausea, diarrhoea; Rare: Swollen tongue.

Skin and subcutaneous tissue disorders: Uncommon: Face swelling, pruritus; Rare: Stevens-Johnson syndrome

Renal and urinary disorders: Rare: Urinary retention.

Reproductive system and breast disorders: Rare: Gynaecomastia.[§]

General disorders and administration site conditions: Uncommon: Malaise.

[§]Adverse drug reaction frequency estimated using “The Rule of 3”.

4.9 Overdose

Signs, Symptoms and Laboratory Findings of Acute Overdosage in Humans

In the postmarketing experience, the most commonly reported adverse events observed with pregabalin when taken in overdose include reduced consciousness, depression/anxiety, confusional state, agitation, and restlessness. Seizures and heart block have also been reported. Deaths have been reported in the setting of lone LYRICA overdose and in combination with other CNS depressants.

Treatment or Management of Overdose

There is no specific antidote for overdose with pregabalin. If indicated, elimination of unabsorbed drug may be attempted by emesis or gastric lavage; observe usual precautions to maintain the airway. General supportive care of the patient is indicated including monitoring of vital signs and observation of the clinical status of the patient. Contact a Certified Poison Control Center for up-to-date information on the management of overdose with pregabalin.

Pregabalin can be removed by hemodialysis. Standard hemodialysis procedures result in significant clearance of pregabalin (approximately 50% in 4 hours).

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antiepileptics , ATC code: N03A (proposed)

The active substance, pregabalin, is a gamma-aminobutyric acid analogue ((S)-3-(aminomethyl)-5-methylhexanoic acid).

Mechanism of action

Pregabalin binds with high affinity to the alpha2-delta site (an auxiliary subunit of voltage-gated calcium channels) in central nervous system tissues. Although the mechanism of action of pregabalin has not been fully elucidated, results with genetically modified mice and with compounds structurally related to pregabalin (such as gabapentin) suggest that binding to the alpha2-delta subunit may be involved in pregabalin's anti-nociceptive and antiseizure effects in animals. In animal models of nerve damage, pregabalin has been shown to reduce calcium-dependent release of pro-nociceptive neurotransmitters in the spinal cord, possibly by disrupting alpha2-delta containing-calcium channel trafficking and/or reducing calcium currents. Evidence from other animal models of nerve damage and persistent pain suggest the anti-nociceptive activities of pregabalin may also be mediated through interactions with descending noradrenergic and serotonergic pathways originating from the brainstem that modulate pain transmission in the spinal cord.

While pregabalin is a structural derivative of the inhibitory neurotransmitter gamma-aminobutyric acid (GABA), it does not bind directly to GABAA, GABAB, or benzodiazepine receptors, does not augment GABAA responses in cultured neurons, does not alter rat brain GABA concentration or have acute effects on GABA uptake or degradation. However, in cultured neurons prolonged application of pregabalin increases the density of GABA transporter protein and increases the rate of functional GABA transport. Pregabalin does not block sodium channels, is not active at opiate receptors, and does not alter cyclooxygenase enzyme activity. It is inactive at serotonin and dopamine receptors and does not inhibit dopamine, serotonin, or noradrenaline reuptake.

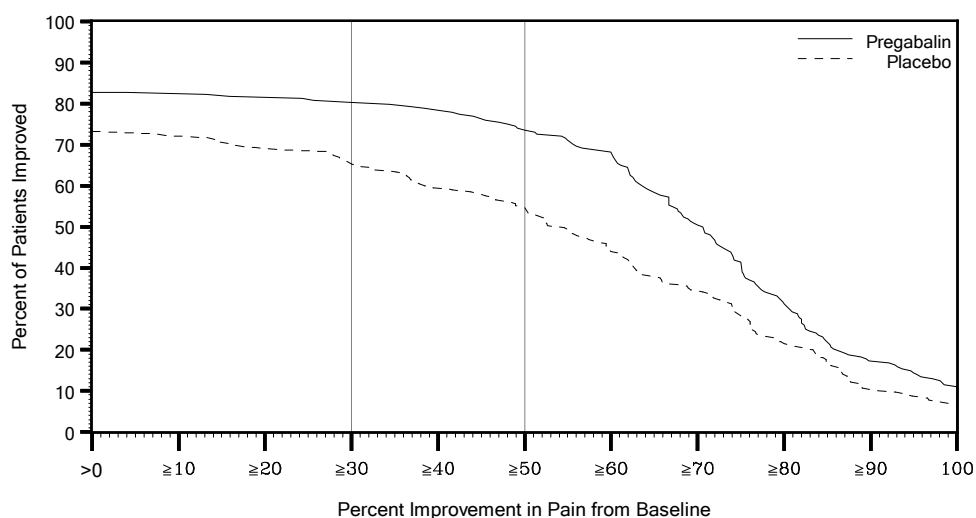
Clinical experience

Management of Postherpetic Neuralgia (Study PHN CR)

Support for efficacy of LYRICA CR for the management of PHN and diabetic peripheral neuropathy (DPN) was based on the efficacy of LYRICA for these indications along with an adequate and well-controlled study in adults with PHN. This 19-week randomized withdrawal study compared daily doses of LYRICA CR 82.5 mg, 165 mg, 247.5 mg, 330 mg, 495 mg, or 660 mg with placebo. Those enrolled were required to have pain present for more than 3 months after healing of the herpes zoster skin rash and a baseline pain score of greater than or equal to 4 on the numeric rating scale (NRS)-Pain (assessed over a 1 week recall period). The baseline mean pain scores were 6.83 for LYRICA CR-treated patients vs. 6.85 for placebo-treated patients. A total of 82.4% of patients completed the single-blind phase of the study. Patients were considered responders if they had at least a 50% reduction in pain in the single-blind phase. Those who responded to treatment were then randomized in the double-blind phase to treatment with either the LYRICA CR dose achieved in the single-blind phase or placebo. Patients were treated for up to 3 months following randomization. A total of 87.5% of LYRICA CR-treated patients and 78% of placebo-treated patients completed the double-blind phase of the study.

LYRICA CR treatment demonstrated statistically significant improvement in the endpoint change in mean pain score from baseline compared to placebo. For a range of levels of improvement in pain intensity from baseline to study endpoint, Figure 1 shows the fraction of patients achieving that degree of improvement. The figure is cumulative, so that patients whose change from baseline is, for example, 50%, are also included at every level of improvement below 50%. Patients who did not complete the study were assigned 0% improvement. In the LYRICA CR group, 79.8% of subjects achieved at least a 30% improvement and 73.6% at least 50% improvement in pain intensity. In the placebo group, 64.9% of subjects achieved at least a 30% improvement and 54.6% at least a 50% improvement in pain intensity.

Figure 1. Percent of Patients Achieving Various Levels of Improvement in Pain Intensity (N=413)



Management of Fibromyalgia (Study FM CR)

A double-blind, placebo-controlled, randomized withdrawal trial of LYRICA CR in adults with fibromyalgia failed to demonstrate efficacy.

Adjunctive Therapy for Adult Patients with Partial Onset Seizures

A double-blind, placebo-controlled, randomized trial of LYRICA CR as adjunctive therapy in adults with partial onset seizures failed to demonstrate efficacy.

5.2 Pharmacokinetic properties

LYRICA CR has linear pharmacokinetics with dose-proportional increases in maximum plasma concentration (C_{max}) and area under the plasma concentration-time curve (AUC) from 82.5-660 mg/day. Following repeated administration, steady state is achieved within approximately 48-72 hours.

LYRICA CR administered once daily following an evening meal has equivalent AUC and lower C_{max} relative to a comparative dose of LYRICA administered without food twice daily (Table 5). Variability in C_{max} and AUC for LYRICA CR is less than or equal to 25%.

Table 5. Steady-State Pharmacokinetics for LYRICA CR™ 165 mg Once Daily and LYRICA™ 75 mg Twice Daily

	LYRICA CR™ Once Daily	LYRICA™ BID
N	24	24
C _{max} (µg/mL)	2.0 (17)	3.2 (21)
T _{max} (h)	8.0 (5.0 – 12.0)	0.7 (0.7 – 1.5)
AUC ₂₄ (µg•h/mL)	29.4 (17)	31.5 (18)
C _{min} (µg/mL)	0.44 (24)	0.59 (25)

Note: Geometric mean (%CV) for AUC₂₄, C_{max}, C_{min}; median (range) for T_{max}.

Abbreviations: AUC₂₄=area under the curve over 24 hours; BID=every 12 hours; C_{max}=peak concentrations; C_{min}=minimum concentrations; N=Number of subjects ; T_{max}=time to peak concentrations.

Absorption

Pregabalin is absorbed from the small intestine and proximal colon. LYRICA CR absorption is linear and dose proportional.

The bioavailability of LYRICA CR is reduced if taken on an empty stomach. The AUC is approximately 30% lower when LYRICA CR is administered fasted relative to following an evening meal.

When LYRICA CR is administered following a 600 to 750 calorie (50% carbohydrates, 20% protein, 30% fat) evening meal, peak plasma concentrations occur within approximately 8 to 10 hours and AUC is approximately 93% to 97% relative to a comparative dose of LYRICA™. The rate and extent of LYRICA CR absorption is similar when administered following a 400 to 500 calorie, 30% fat or an 800 to 1000 calorie, 15%, 30%, or 50% fat evening meal.

When LYRICA CR is administered following an 800 to 1000 calorie (50% carbohydrates, 20% protein, 30% fat) morning meal, peak plasma concentrations occur within approximately 12 hours and AUC is 99% relative to a comparative dose of LYRICA. AUC decreases approximately 13% to 25% when LYRICA CR is administered following a 400 to 500 calorie or 600 to 750 calorie (50% carbohydrates, 20% protein, 30% fat) morning meal relative to the 800 to 1000 calorie meal, while C_{max} remains the same.

Distribution

Pregabalin does not bind to plasma proteins. The apparent volume of distribution of pregabalin following oral administration is approximately 0.5 L/kg. Pregabalin is a substrate for system L transporter which is responsible for the transport of large amino acids across the blood brain barrier. Although there are no data in humans, pregabalin has been shown to cross the blood brain barrier in mice, rats, and monkeys. In addition, pregabalin has been shown to cross the placenta in rats and is present in the milk of lactating rats.

Metabolism

Pregabalin undergoes negligible metabolism in humans. Following a dose of radiolabeled pregabalin, approximately 90% of the administered dose was recovered in the urine as unchanged pregabalin. The N-methylated derivative of pregabalin, the major metabolite of pregabalin found in urine, accounted for 0.9% of the dose. In preclinical studies, pregabalin (S-enantiomer) did not undergo racemization to the R-enantiomer in mice, rats, rabbits, or monkeys.

Elimination

Pregabalin is eliminated from the systemic circulation primarily by renal excretion as unchanged drug with a mean elimination half-life of 6.3 hours in subjects with normal renal function. Mean renal clearance was estimated to be 67.0 to 80.9 mL/min in young healthy subjects. Because pregabalin is not bound to plasma proteins this clearance rate indicates that renal tubular reabsorption is involved. Pregabalin elimination is nearly proportional to CL_{Cr} [see Dosage and Administration (2.5)].

Pharmacokinetics in special patient groups

Elderly (over 65 years of age)

Pregabalin oral clearance tended to decrease with increasing age. This decrease in pregabalin oral clearance is consistent with age-related decreases in CL_{Cr}. Reduction of pregabalin dose may be required in patients who have age-related compromised renal function (see **Section 4.2 Posology and method of administration**).

Gender

Population pharmacokinetic analyses of the clinical studies showed that the relationship between daily dose and LYRICA CR drug exposure is similar between genders.

Race/Ethnicity

In population pharmacokinetic analyses of the clinical studies of LYRICA and LYRICA CR, the pharmacokinetics of pregabalin were not significantly affected by race (Caucasians, Blacks, and Hispanics).

Renal impairment

Pregabalin clearance is nearly proportional to CL_{Cr}. Dosage reduction in patients with reduced renal function is necessary. Pregabalin is effectively removed from plasma by hemodialysis. Following a 4-hour hemodialysis treatment, plasma pregabalin concentrations are reduced by approximately 50%. For patients on hemodialysis, treatment with LYRICA CR is not recommended (see **Section 4.2 Posology and method of administration**).

Breast-feeding mothers

The pharmacokinetics of 150 mg pregabalin given every 12 hours (300 mg daily dose) was evaluated in 10 lactating women who were at least 12 weeks postpartum. Lactation had little to no influence on pregabalin pharmacokinetics. Pregabalin was excreted into breast milk with average steady-state concentrations approximately 76% of those in maternal plasma. The estimated average daily infant dose of pregabalin from breast milk (assuming mean milk consumption of 150 mL/kg/day) was 0.31 mg/kg/day, which on a mg/kg basis would be approximately 7% of the maternal dose.

5.3 Preclinical safety data

In conventional safety pharmacology studies in animals, pregabalin was well-tolerated at clinically relevant doses. In repeated-dose toxicity studies in rats and monkeys CNS effects were observed, including hypoactivity, hyperactivity and ataxia. An increased incidence of retinal atrophy commonly observed in aged albino rats was seen after long-term exposure to pregabalin at exposures ≥ 5 times the mean human exposure at the maximum recommended clinical dose.

Teratogenicity

Pregabalin was not teratogenic in mice, rats or rabbits. Foetal toxicity in rats and rabbits occurred only at exposures sufficiently above human exposure. In pre-natal/post-natal toxicity studies, pregabalin induced offspring developmental toxicity in rats at exposures > 2 times the maximum recommended human exposure.

Mutagenicity

Pregabalin is not genotoxic based on results of a battery of *in vitro* and *in vivo* tests.

Carcinogenicity

Two-year carcinogenicity studies with pregabalin were conducted in rats and mice. No tumors were observed in rats at exposures up to 24 times the mean human exposure at the maximum recommended clinical dose of 600 mg/day. In mice, no increased incidence of tumors was found at exposures similar to the mean human exposure, but an increased incidence of haemangiosarcoma was observed at higher exposures. The non-genotoxic mechanism of pregabalin-induced tumor formation in mice involves platelet changes and associated endothelial cell proliferation. These platelet changes were not

present in rats or in humans based on short-term and limited long-term clinical data. There is no evidence to suggest an associated risk to humans.

In juvenile rats the types of toxicity do not differ qualitatively from those observed in adult rats. However, juvenile rats are more sensitive. At therapeutic exposures, there was evidence of CNS clinical signs of hyperactivity and bruxism and some changes in growth (transient body weight gain suppression). Effects on the estrus cycle were observed at 5-fold the human therapeutic exposure. Neurobehavioral/cognitive effects were observed in juvenile rats 1-2 weeks after exposure >2 times (acoustic startle response) or >5 times (learning/memory) the human therapeutic exposure. Reduced acoustic startle response was observed in juvenile rats 1-2 weeks after exposure at >2 times the human therapeutic exposure. Nine weeks after exposure, this effect was no longer observable.

Juvenile Animal Toxicity Data

In studies in which pregabalin (50 to 500 mg/kg) was orally administered to young rats from early in the postnatal period (Postnatal Day 7) through sexual maturity, neurobehavioral abnormalities (deficits in learning and memory, altered locomotor activity, decreased auditory startle responding and habituation) and reproductive impairment (delayed sexual maturation and decreased fertility in males and females) were observed at doses greater than or equal to 50 mg/kg. The neurobehavioral changes of acoustic startle persisted at greater than or equal to 250 mg/kg and locomotor activity and water maze performance at greater than or equal to 500 mg/kg in animals tested after cessation of dosing and, thus, were considered to represent long-term effects. The low effect dose for developmental neurotoxicity and reproductive impairment in juvenile rats (50 mg/kg) was associated with a plasma pregabalin exposure (AUC) approximately equal to human exposure at the maximum recommended dose of 660 mg/day. A no-effect dose was not established.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Kollidon® SR (contains Polyvinyl Acetate, Povidone, Sodium Lauryl Sulfate, Silica), Crospovidone, Polyethylene Oxide, Carbomer Homopolymer, Magnesium Stearate

In 82.5 mg tablet: Opadry® II Complete Film Coating System 85F105027 Blue; contains Polyvinyl Alcohol, Talc, Titanium Dioxide, Macrogol 3350, Indigo Carmine Aluminium Lake

In 165 mg tablet: Opadry® II Complete Film Coating System 85F170009 Beige; contains Polyvinyl Alcohol, Talc, Titanium Dioxide, Macrogol 3350, Iron Oxide Yellow, Iron Oxide Red

6.2 Incompatibilities

Incompatibilities were either not assessed or not identified as part of the registration of this medicine.

6.3 Shelf life

Observe “Expiry date” imprint on bottle.

6.4 Special precautions for storage

Store below 25°C. Keep in a cool, dry place away from sunlight.

6.5 Nature and contents of container

LYRICA® CR 82.5 mg:
Bottles of 30's modified-release tablets.

LYRICA® CR 165 mg:
Bottles of 30's modified-release tablets.

6.6 Special precautions for disposal of a used medicinal product or waste materials derived from such medicinal product and other handling of the product

No special requirements.

7. MANUFACTURER

Pfizer Manufacturing Deutschland GmbH
Mooswaldallee 1
Freiburg Im Breisgau
Germany

Date of Revision: 23 Oct 2025

PI-LYRICACR-1025